

Shutter, Aperture and ISO:

There are 3 important things you will need to focus on with shooting in manual mode camera settings: the shutter speed, the aperture and the ISO. Shooting in manual mode is so important because it allows you to have full control over what your images look like. With my personal style, I love to shoot quite sharp but with a nice depth of field. So I typically tend to shoot at a low aperture of around f/1.4 to get that unique style.

I also love shooting in manual mode because it makes editing in Adobe Lightroom a breeze when all of my photos look exactly the same and my camera settings aren't shifting all over the place in between every photo. While I never shoot using auto mode camera settings now, I did use these settings when I first began my photography journey.

I found that when I turned on auto mode camera settings in the past, the settings for my photos ended up being pretty all over the place, which made my job pretty difficult when editing in Lightroom. For example, my aperture might have jumped from f/2.5 all the way up to f/9 just from moving my camera in a slightly different direction. You have no control over what your camera will do, and this limits you a lot in the photo editing process.

Basic Camera Settings:

I'm going to give you a quick rundown over what the shutter speed, aperture and ISO are. Essentially, the shutter is the mirror inside the camera flipping open and close every time you take a photo. On the Canon 5D Mark iii, you can have it as fast as 1/8000, and if you want to do a long exposure, you can have it as low as 30 seconds (so the mirror will stay open for 30 whole seconds). The aperture are the blades inside your lens. So if you have it open at 1.4, the blades will be wide open, resulting in a lot of light coming in to your photo and a really wide depth of field (which is how you get the unique style of bokeh and a blurry background). If you put your aperture settings higher, say at 1.8, the blades close so that you get less light in the photo with a result of more sharpness as well.

And then we have ISO which is your sensor sensitivity. I always leave my ISO at the lowest possible setting which is 100. You can also put your ISO up to very high settings like 100,000, but you typically only need to do this for night photography or if it's really dark in your shooting environment. However, keep in mind that the higher your ISO is, the more grainy your images will be. On the 5D Mark iii, I normally don't like to go past 6400. I find that anything beyond that is a little bit too grainy for my style. So I like to generally keep it between 6400 and 100 when I'm shooting.

Basic Photography Tips

So jumping into actually shooting on manual mode. I know that it can be a bit daunting at first if you've never done it before or have always shot with auto settings turned on. However, I promise that once I explain the way I do things and you go out and practice this, it honestly becomes second nature in no time.

Shooting in manual mode really doesn't take any time away from the photoshoot- it really only makes you a stronger photographer in the long run.

Normally when I'm shooting personal work, I like to leave my aperture wide open— somewhere between f/1.4 (the lowest value my lens is capable of) and 2.5, because I personally just love the style of that bokeh in the background and nice depth of field (blurry look). If I were shooting for paying clients, I would still like to keep my depth of field style on my photos, but I would want to make sure the images were very sharp. So in that case, I would typically shoot at f/2.8 or f/3.5. If I'm shooting landscape photography (as opposed to portrait photography), I like to move my aperture settings up to around f/5.6 to make sure that everything is sharp in the image.

Shooting In Manual Mode:

With that in mind, the first thing I like to do when shooting in manual is selecting what aperture I want to be doing the shoot at. That is typically where I will leave my aperture for the entire day unless the lighting changes dramatically. For example, if it gets really dark, I want to make sure my aperture is at the lowest possible setting, therefore letting in the most light possible (in my case, my lens is capable of a f/1.4 aperture, so that is what I would choose for dark lighting).

The next thing I want to do is move my ISO camera settings down to 100. And once again, if I'm shooting out in the sunlight, ISO 100 is just where it will stay the entire time I shoot. So now you have 2/3 settings that are pretty much set and will not change for the remainder of your shoot.

So, the last thing you will need to set on manual is the shutter speed, and that is the setting that is the most important. So what I typically do is point my camera in the direction that I will be taking the photo, and taking a look at my evaluation meter (the lighting meter that's in the top of the camera). Basically I will just move the meter until I get the little line in the middle). Then I will take a test shot, have a look at the photo, and more likely than not, the exposure will be correct. However, you can make some changes as needed. For example, if I look at the image and it looks a bit dark, then I can just move my shutter down a little bit, and by then, I'll be ready to shoot.

So I basically do that for each location and each different lighting situation I have for a shoot. So if that all sounds pretty easy, that's because it kind of is! Honestly, you just need to practice a little bit, take your time and think about what you're doing.

Best Camera Settings— Outdoor Photography

One of the first camera settings that should be changed for outdoor, bright light photography is the ISO settings. An ISO setting of 100 is going to be the most optimal, so anytime you can swing it— you definitely should. A lower ISO means that you should be shooting in an environment with bright lighting, as the setting will be letting less light into the camera.

Canon Settings:

Below, I've included a set of photos that my cousin and I recently took during our Europe trip: I'm including the camera settings used below so that you can get a better idea of why using

manual camera settings can offer much more consistency among a photo set (and understand photography basics a little better for factors such as shooting environment and lighting).

Aperture: f/2.5; with a relatively low f-stop (f/1.4 is the lowest that my lens is capable of), my cousin and I were able to achieve a bit of a blurry background, while still retaining enough detail in the building and flags behind me.

ISO: 100; this is what I always set my ISO at if I'm shooting in optimal lighting (i.e. bright light/outdoors). The lower your ISO, the sharper your images will be. If it were getting darker outside (maybe 5 or 6 p.m.), I might have to start bumping my ISO up a bit— somewhere around 200-300 ISO should work fine unless the sun has already set.

Shutter Speed: 1/1200; because my cousin and I were shooting with a higher shutter speed, she was able to capture my movement (and it was a particularly windy day which added even more motion to the photos). If it was later in the evening/ getting darker outside, we would have had to shoot using a lower shutter speed. This would have made it more difficult to capture the motion that we were able to get with this set (unless we were using a tripod).

Final Notes/ My Rules:

To sum things up, I have a few rules I typically like to go by when shooting in manual mode.

I always keep my ISO at 100 if I'm shooting in the daytime. There really isn't any need to bring that up more, unless it breaks the following rule...

I don't like to bring the shutter less than 1/250, if possible. This is to eliminate movement blur in my photography. In my personal style, I like for my photos to look a bit like a movie still, so I don't like movement blur in my imagery (but there are certain shooting environments where I will break this rule, like in the photo set that I included as an example above).

Why shoot in Manual Mode?

Here's the thing— you don't know what you don't know.

via GIPHY

Maybe you have no idea why you're even supposed to shoot in manual mode.. but your photographer friend Sally said you need to figure it out ASAP if you want top-notch photos. So now you're here..

Well good news for you— I'm nicer & more helpful than your friend Sally. :)

Lets put it this way. By using manual camera settings, we can have complete control over the way our images turn out. And more specifically, we can control how much LIGHT we let into our camera:

If we have too much light entering the camera, our photo will end up being overexposed (meaning way too bright).

If we don't have enough light entering the camera, our photo will be underexposed (meaning too dark).

And we need to have a properly exposed photo if we want that perfect photo for Instagram, duh! And thats exactly what we can achieve by shooting with manual settings— instead of letting our camera dictate how much light comes in, we get to call the shots on everything.. including how Instagram-worthy our photo turns out. ;)

The Exposure Triangle

One thing we should always aim for with photography is to get our image properly exposed—and there are 3 essential settings we need to focus on to get the proper exposure:

Shutter speed

Aperture

ISO

I'll go in way more detail on each setting individually... but its important to note that our overall goal is to get these 3 in balance until our image is properly exposed. And we can do this via the light meter on our camera's viewfinder (see below).

Balance Exposure with the DSLR Light Meter

All cameras have a light meter; this is the little number line displayed on top of the camera body or in the viewfinder. The meter won't display immediately, but it'll appear as soon as you half-press the shutter button to focus in on your subject.

Here's what it'll look like for both Canon & Nikon users:

Canon: -2...1...0...1...2+

Nikon: +2...1...0...1...-2

The light meter is basically an indicator that tells us where our exposure is in relation to the 0.

If we're to the left of 0, our photo will be underexposed (not bright enough). If we're to the right of 0, our photo will be overexposed (too bright). The overall goal is to be right at the center (0).. this means that our image will be "properly exposed".

Personally, I always aim to be sliiiiightly to the left of 0, not right at the center. Its much easier to correct for a slightly underexposed image than it is an overexposed image. Thats just me personally though! I know many people like super bright edits so they go more to the right. Just aim to not be toooo far off 0 or you might not be able to save the photo at all!

What is the Shutter Speed Setting?

Shutter speed is the speed at which your camera shutter opens & closes (think: how fast the shutter *clicks* to take your photo). By manually controlling for shutter speed, we can regulate how quickly light enters the camera & how much movement (or motion blur) is captured in our final image.

The actual shutter speed value itself is denoted by fractions of a second. For example:

1 sec would be considered a "slow" shutter speed

1/2000 sec would be considered a "fast" shutter speed

The best way to think of shutter speed is through its ability to freeze or blur motion in your photo:

The longer your shutter remains open (aka using a slow shutter speed like 1 sec), the more light & movement will be captured in your image. Andddd that means your photos can end up super blurry if you're shooting handheld & not with a tripod.

The less time a shutter remains open (aka using a fast shutter speed like 1/2000 sec), the less light & movement will be captured in your image. But thats where it can sound a bit confusing: when I say "less movement will be captured", its actually a good thing. It means that I can freeze a subject in motion perfectly and our photo won't turn out blurry.

So you know all those cool hair-flip pictures or creative action shots where every single detail is perfectly in focus & you're just sitting there thinking, "how the heck is this photo even possible?".

Ya... been there, wondered that.

The secret? Using a fast shutter speed. By using a fast shutter speed (lets say 1/700 sec) you can capture SO much motion in your photos.

Lets just break it down for a sec with a couple examples. *I'm going to assume that we're using a low ISO & aperture of f/2.5 for both (we'll get into ISO & Aperture later).*

1. So lets just say you're outside taking a photo of someone flipping their hair (in daylight hours). You're using a SLOW shutter speed (i.e. 1/60 sec). Would you be able to capture the action?

Using a fast shutter speed

Here's another practical example of using a fast shutter speed. When my cousin Kasey (she's the real Instagram photo queen) joined me in my Europe travels this past March, we were testing out #AllTheThings to get those perfect Instagram shots: different camera settings, lighting situations & shooting environments.

One thing we consistently aimed for with our photoshoots, however, was capturing movement — whether it be through shaking our hair back & forth, doing leaps in the air (all those years in dance finally paid off!) or simply standing in place with moving cars whizzing by in the background. And by using a fast shutter speed, we were able to capture some really cool effects in our images! If we had used a slow shutter speed for these, we wouldn't have been able to freeze the motion like we did.

I included a few of our Paris photos below for reference — the shutter speed values on these ranged from 1/800 to 1/2500 sec.

Using a slow shutter speed

Using a slow shutter speed wouldn't be very effective for sports or action photography where the main goal is to capture (or "freeze") your subject's movement.

Using a slow shutter speed can, however, create a beautiful effect for night photography — one of my high school friends, Ernie Carr, has great examples of these effects on his Instagram photography account which showcases crispy, Disney images (and a TON of the images are taken at night).

P.S., I promise I'm not being biased here but Ernie is SERIOUSLY talented and shares SO much knowledge on his Insta account— I highly suggest checking him out for photography tips, inspiration... or just to stare at his amazing content in awe.

Again, its important to note that a slower shutter speed will be more sensitive to camera movements (aka motion blur caused by holding the camera in hand).

This can, however, be prevented easily through the use of a sturdy tripod! Here's a few of the best budget tripods I recommend:

Best Shutter Speed Settings

There's no right or wrong answer here... but it is helpful to keep a general set of rules for your camera settings (which can always be modified based on your shooting environment).

My personal rule for shutter speed is to never go below 1/100 of a second if I'm holding the camera in my hand (but we all make mistakes, especiallllly if we're in a rush).

Other than that, here's a few common shutter speeds for different situations:

Basic street photography: 1/250 sec, 1/500 sec

Landscape photography (without tripod): 1/200 sec

Landscape photography (with a tripod): 1 sec

Night photography: 1/4 sec, 1/8 sec or even something drastic like 15-20 seconds (but again, you'll need a tripod for this).

Shutter Speed Cheat Sheet

When we use a slow shutter speed (ie. 1 sec), we increase the brightness

BUT in doing so, we also run the risk of having a BLURRY PHOTO (if a tripod isn't used)

When we use a fast shutter speed (ie. 1/2000 sec), we decrease the brightness

BUT we're able to capture quick movement in our photos like someone running, water dripping, someone flipping their hair etc.

What is the Aperture camera setting?

The more technical Aperture Definition

If you want the technical jargon, aperture is the diameter of the entrance pupil of the lens through which light travels into the camera. And focal ratio is a measurement of the size of the aperture, often referred to as "f-stop".

.....excuse me, what?

Fallon's Aperture Definition

Alright let me explain that a little better...

Aperture is basically a setting which dictates how much light you're letting into your camera when you open & close your lens. Let me repeat— your lens. That's the key thing to note here— aperture is alllll about the lens you're shooting with. And it's important to understand that:

A larger aperture (or opening of the lens) will let MORE light into the camera (making our photo brighter).

A smaller aperture (or closing of the lens) will let LESS light into the camera (making our photo darker)

F-stop goes hand-in-hand with aperture.. and is super important to understand as well. The f-stop is a measurement (aka the actual number) which is indicated as a mark on our lens. These values typically range from f/1.2 to f/22.

Again, the f-stop is all going to depend on the LENS you're shooting with. If you're shooting with the kit lens that came with your camera (which is totalllly fine), you might be a little limited in changing your f/stop. If you've researched & invested in a professional lens, you might have more flexibility.

I'll give you a personal example to lay this out: I'm currently shooting (most) of my photos with a Canon 35mm f/1.4 lens Technicallllly... this would be described as having a "maximum aperture of 1.4". But thats where things get a little confusing:

A larger (or "maximum") aperture is going to be indicated by a SMALLER NUMBER (I.e. f/1.4)

A smaller aperture is going to be indicated by a LARGER NUMBER (I.e. f/22).

This basically means that I have a little more flexibility with my lens because it can open up all the way to f/1.4.

But I swear... its not nearly as confusing when you actually put it into practice! The same advice sticks for any of the camera settings tips/advice in this article. Everything really starts to come together once you actually test it out & see what works!

How to get depth of field or blurry background
Maybe even more important to you...

When we change our aperture (or f-stop) using manual settings, we're also affecting the depth of field... AKA that "blurry background" effect everyone loves so much.

This is where having a low f/stop comes in handy:

By decreasing the f-stop, we'll create a shallow depth of field— this means that less of our overall image will be in focus. Instead, the focus will be centered in on one subject (like in the photo below). You might want this effect for portraits or close-up, detail shots.

By increasing the f-stop, we'll create a deeper depth of field— this means that more of our overall image will be in focus. You might want this effect for landscape shots or any photos where you need to get the FULL image in focus rather than just one subject.

Aperture Cheat Sheet & Key Takeaways

Aperture (or f-stop) is the size of the hole in your lens that allows light to enter your camera. By controlling for this setting manually, we can:

Open the lens to make our photo brighter (using a smaller f/stop such as f/1.4)

Close the lens to make our photo darker (using a larger f/stop such as f/8)

So by using a smaller f/stop (ie f/1.4), you'll let in MORE light & get that "blurry background" effect (great for portrait photos). By using a larger f/stop (ie. f/8), you'll let in LESS light & your ENTIRE image will be sharp & in focus (great for landscape photos).

What is the ISO camera setting?

The more technical ISO Definition

If we're getting technical here.. ISO (which is governed by the International Organization for Standardization) is the sensitivity of a camera's sensor to light; it allows more light to be captured without sacrificing other camera settings in order to get your exposure, or light meter, in balance (i.e. by decreasing shutter speed or increasing aperture).

Fallon's ISO Definition

ISO is just a fancy way of saying "increasing the brightness". BUT the key thing to note here is that by increasing your ISO, you're also going to increase the amount of GRAIN or "digital noise" in your final image.

Andddd let's be honest, that's not exactly the best look. Even if you're into the whole "grainy, vintage style" that's so trendy these days (which is pulled off very well by NYC-based Instagram Influencer & photographer Tezza Barton) you'll just wanna take note: this is NOT the same kind of grain.

Those vintage-style Instagram photos have "artificial grain" added to them using the grain slider in Adobe Lightroom. So, let's clear things up. The grain you see in your image after increasing the ISO is not going to provide you with a trendy, vintage look.

That's why I like to think of ISO as the absolute last resort in changing manual camera settings — and your goal should be to keep this value as LOW as possible. To give you a little more context... ISO is measured along a scale, with common values ranging from 100 to 1600. See below for reference:

Best ISO Camera Settings

Personally, I avoid my ISO setting like the plague and will almost always keep it at 100 until it becomes dark outside. *But with that being said.. this rule has gotten me in some trouble in the past & I'm now being more careful about increasing my ISO when necessary.*

In general... my rule is to keep my ISO setting on 100 unless:

The photoshoot takes place indoors

I'm shooting night photography

The lighting changes drastically (ie. transition from "golden hour" to nighttime, skies get cloudy, etc).

In those cases, you'll want to gradually increase the ISO value until your exposure is balanced out.

ISO Cheat Sheet & Key Takeaways

When we increase ISO, we increase the brightness

BUT in doing so, we also increase the amount of digital noise or GRAIN in our final image

Keep ISO as LOW as possible during DAYLIGHT hours (ie. ISO 100) and increase gradually when it starts to get dark out

Putting Manual Mode in Practice

Alright— lets just use this quick example to break down all the settings we've learned so far.

The photo below was taken at a restaurant in Germany back in 2017 before I knew anything about photography. I'm talking nothing, zip, nada — fully automatic camera settings were used and I was shooting in JPEG format (I'll cover RAW vs JPEG in a separate blog post).

I was fortunately able to work with these settings in Lightroom and get the final edit looking decent.

However, if we take a closer look at the histogram in Lightroom, we can see that our ISO for this image was set at 2000.

And while the effects may not look very drastic from this perspective, if we zoom in...

So what are some ways this high ISO could have been avoided?

Well, while I can't see the DSLR light meter that was used to balance out our exposure, there are a few assumptions that can be made from our settings in the LR histogram.

For starters— its important to note that our ISO would definitely have to be higher than our general rule of 100. How do we know this?

Our image was shot indoors, which typically means we're working with a dark shooting environment and we'll therefore need to increase our ISO past 100.

Despite the already high ISO, our shutter speed wasn't exactly ideal at 1/60 sec. This means that the restaurant lighting was most likely difficult to work with (which is why the camera automatically chose those settings)... and our options in the exposure triangle were limited without using a tripod.

So to avoid this high ISO without sacrificing shutter speed, I would have decreased the f/stop from its current value at f/3.5.

While you won't always have this option (it depends on the lens you're working with), I know that the lens I was using here had a maximum aperture of f/1.8.

So I would have first lowered the f-stop from f/3.5 to f/1.8 — this would have resulted in a stronger focus on the 2 glasses and a more shallow depth of field/ blurry background. But more importantly, it would have given us more room to make adjustments to our shutter speed and ISO values. Because a lower f-stop means a you'll be letting MORE light into your camera!

I could then adjust my shutter speed. The rule I always (try to) follow is to never select a shutter speed slower than 1/100 sec unless I'm using a tripod — this helps avoid motion blur caused by handheld photography. I would therefore increase the shutter speed from 1/60 sec to 1/100 sec.

I would then simply decrease the ISO until the light meter was balanced out at zero and our image was properly exposed.

Final Manual Photography Tips

I think its safe to say that I filled this post with a whole lot of information. So if you made it allllll the way down here, then I have no doubt you'll become an amazing content creator in no time! You're passionate and willing to learn & grow in your craft— thats what really matters. :)

But with that being said, having a general set of rules can definitely help speed up the process! Here's a few of my personal rules— by no means feel obligated to use these; see what works best for your style & roll with it!

My rules for manual settings:

Set ISO at 100 (and gradually increase when it starts getting dark out)

Make sure that shutter speed is never slower than 1/100 sec. For example, 1/125 sec would be fine, but 1/20 sec would not be fine & your photo would end up blurry if handheld. Which leads me to my next point..

If your shutter speed does have to dip below 1/100 sec, make sure you're using a sturdy tripod!

Aperture/f-stop varies. First, it'll depend on the lens you're shooting with (a lot of lenses aren't even capable of going lower than f/4). But it honestly depends on your personal photography

style as well. I struggle with aperture at times because I love the unique effects it can create... but I also like my photos to make bold statements and have enough contrast/detail (which can fade out with a low f-stop). However, in general, my aperture will range from f/2 to f/4, usually sticking somewhere around f/2.5.

Keep Practicing & learning!

Just remember, if this doesn't all "click" instantly... its TOTALLY okay! This stuff takes a long time to master & it truly does just take practice, practice & more practice.

I am not a professional photographer by any means (and would never claim to be) and I have days where I seriously struggle with this stuff too! The thing is, each of the settings can sound relatively simple on paper... until you start facing external factors that you have zero control over (i.e super dark, cloudy skies).

And you're faced with an even bigger challenge when time is of the essence. You can't just whip out the perfect settings on command & call it a day (I wish). It will take some time to get your settings aligned based on the photoshoot location... and some trial/error shots. I can't even begin to tell you how many times I've taken photos for friends & family who didn't have a photography background... only to be met with impatient or confused stares when I was taking too long adjusting my settings.

But with that being said, you will get faster with practice! So just keep learnin' new things (I am seriously always in learning mode) and get out there to see what works/doesn't!